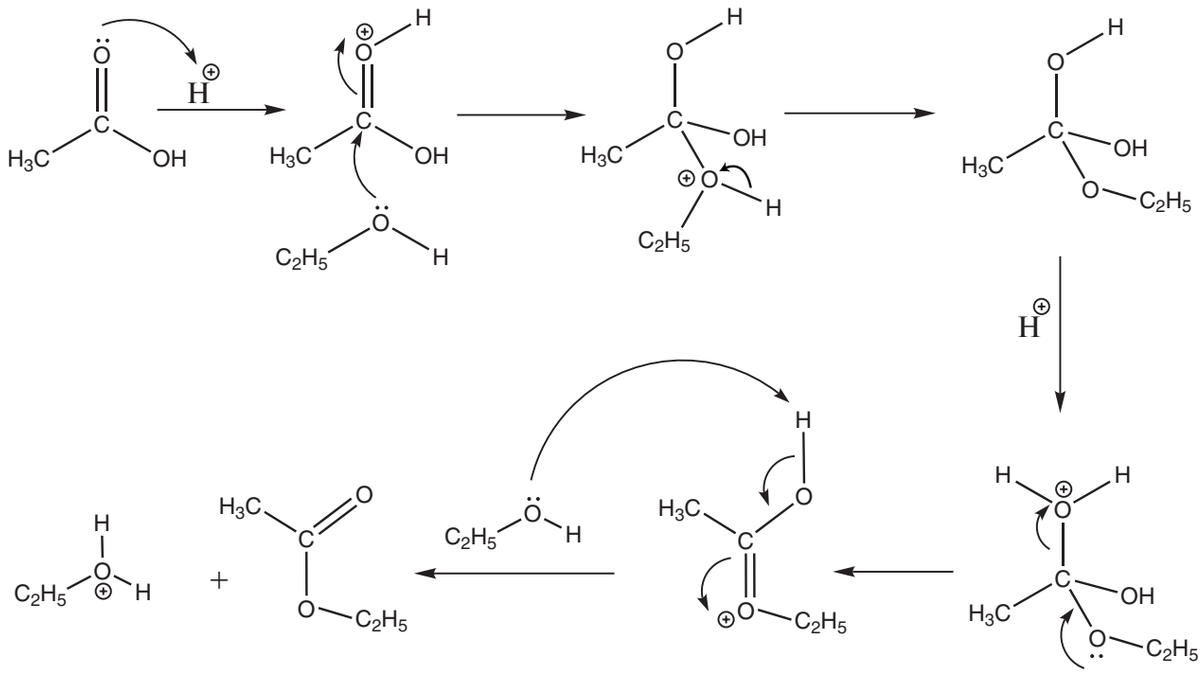
Mechanism of esterification:

The Mechanism of esterification involves the following steps.



C) Reactions involving – COOH group

1) Reduction

i) Partial reduction to alcohols

Carboxylic acids are reduced to primary alcohols by $LiAlH_4$ or with hydrogen in the presence of copper chromite as catalyst. Sodium borohydride does not reduce the – COOH group.

Example

$$CH_3-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[4(H)]{LiAlH_4} CH_3CH_2OH + H_2O$$

ethanoicacid — ethanol

ii) Complete reduction to alkanes

When treated with HI and red phosphorous, carboxylic acid undergoes complete reduction to yield alkanes containing the same number of carbon atoms.

Example

$$CH_3-\overset{\overset{O}{\|}}{C}-OH + 6\,HI \xrightarrow[473\ K]{Red\ P} CH_3-CH_3 + 3I_2 + 2H_2O$$

Acetic acid — Ethane

XII U12-Carbonyl Compounds.indd 174 08-12-2021 22:50:44